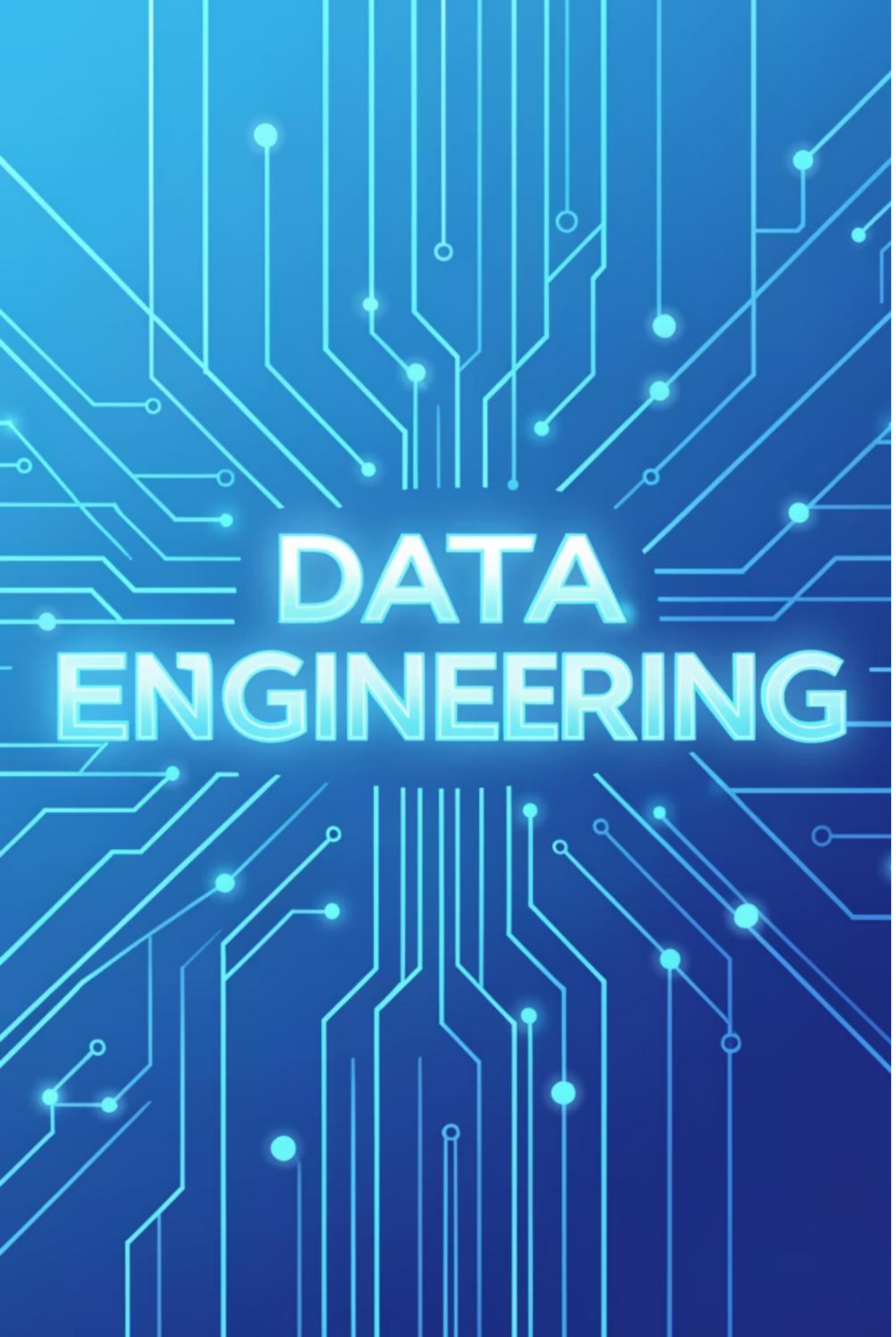


Data Engineering

Spring '26 – DS463

Feb 10, 2026

Lecture 10

Logistics: Your Course Essentials Essentials

Class Timings

Monday (LH6), Tuesday (LH8): 11:30 AM AM – 12:20 PM

Friday (LH8): 11:00 AM – 11:50 AM

Credit Hours

2 hours (Lectures) + 1 hour (Practical (Practical Session)

Office Hours

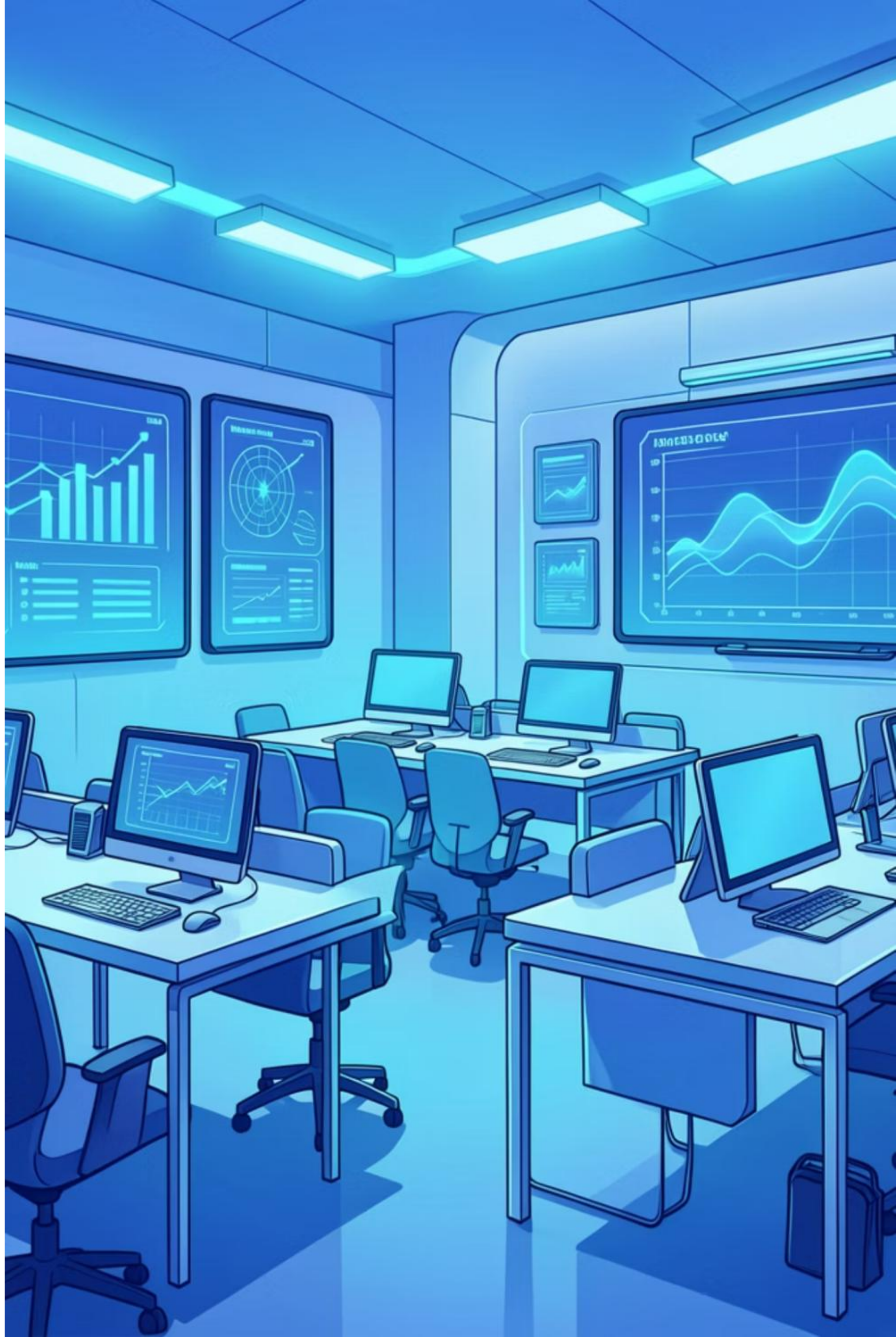
Monday, Tuesday: 10:00 AM – 11:00 AM, AM, S18 2nd Floor NAB or by appointment.

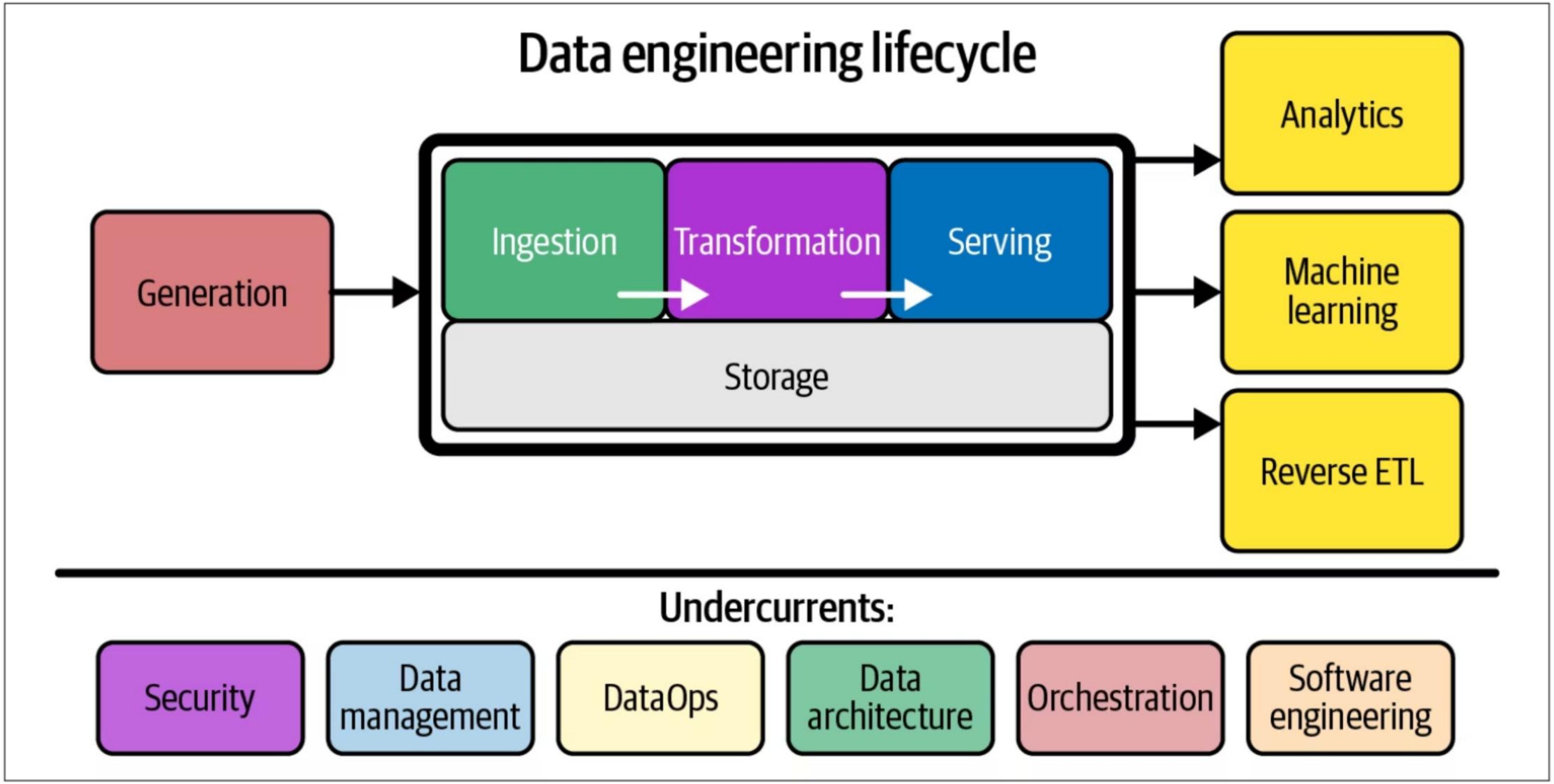
Contact & Support

MS Teams: w5t93yk

Email: farah.saeed@giki.edu.pk

WhatsApp: Reach out for quick queries





Various stages of the lifecycle may repeat themselves, occur out of order, overlap, or weave together in interesting and unexpected ways.

Recap

Transformation

- Querying
- Modelling
- Transformation

Serving
Usecases

- Analytics

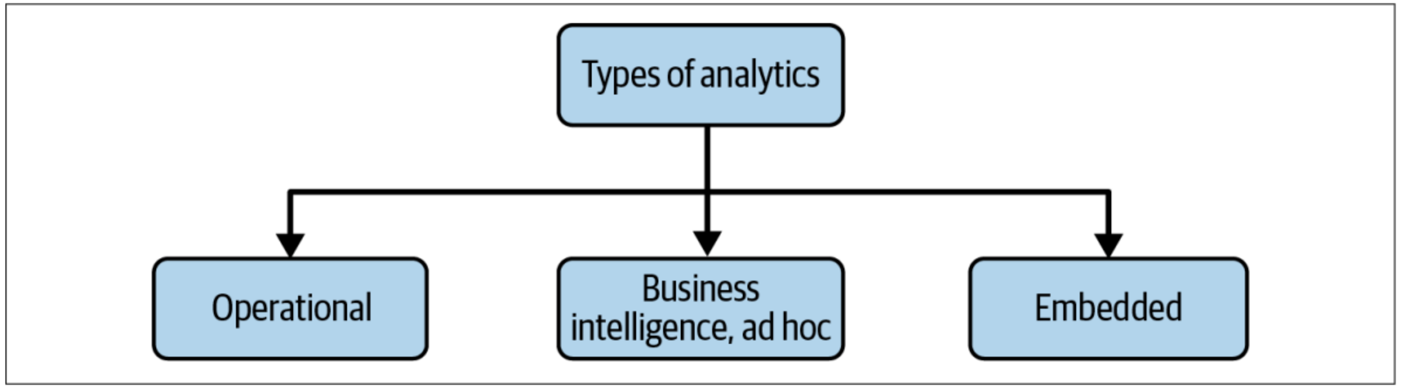


Figure 2-5. Types of analytics

- Machine Learning
- Reverse ETL

What is Data Architecture?

What is Data Architecture?

A data architecture describes **how data is managed**, from collection to transformation, distribution and consumption. – IBM

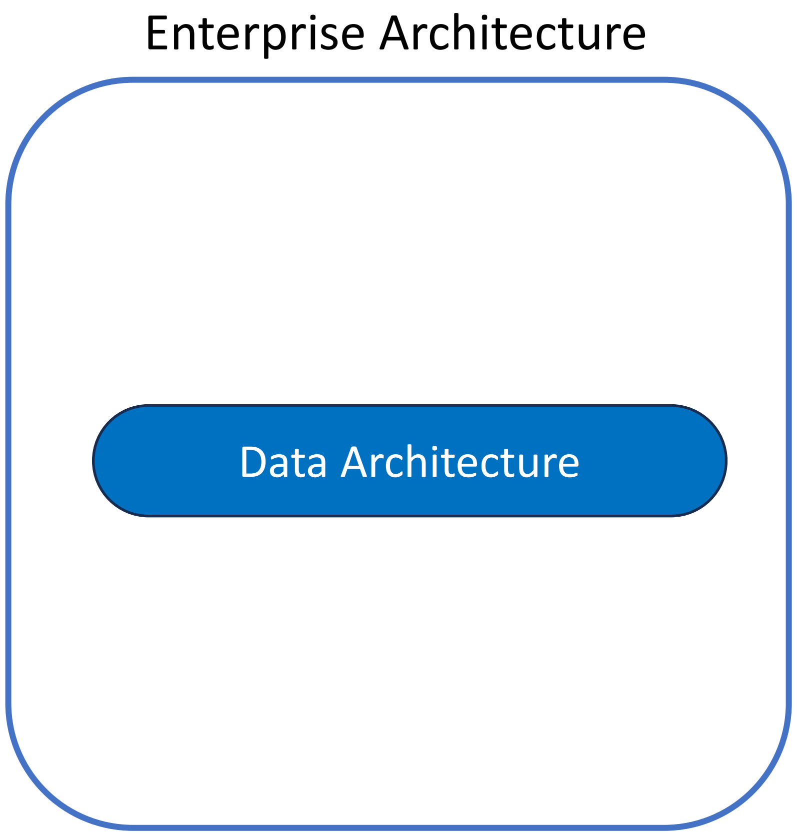
Data architecture is the overarching framework that describes and governs an organization's data collection, management, and usage. – AWS

Enterprise Architecture

Enterprise Architecture

“The design of systems to **support change in an enterprise**, achieved by flexible and reversible decisions reached through careful evaluation of tradeoffs.”

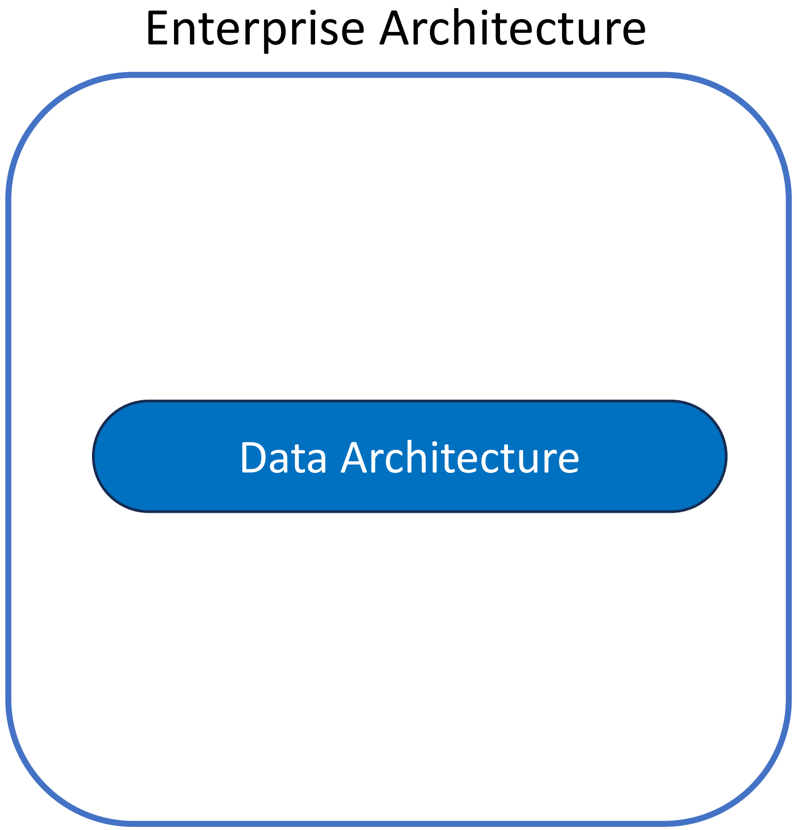
Enterprise Architecture



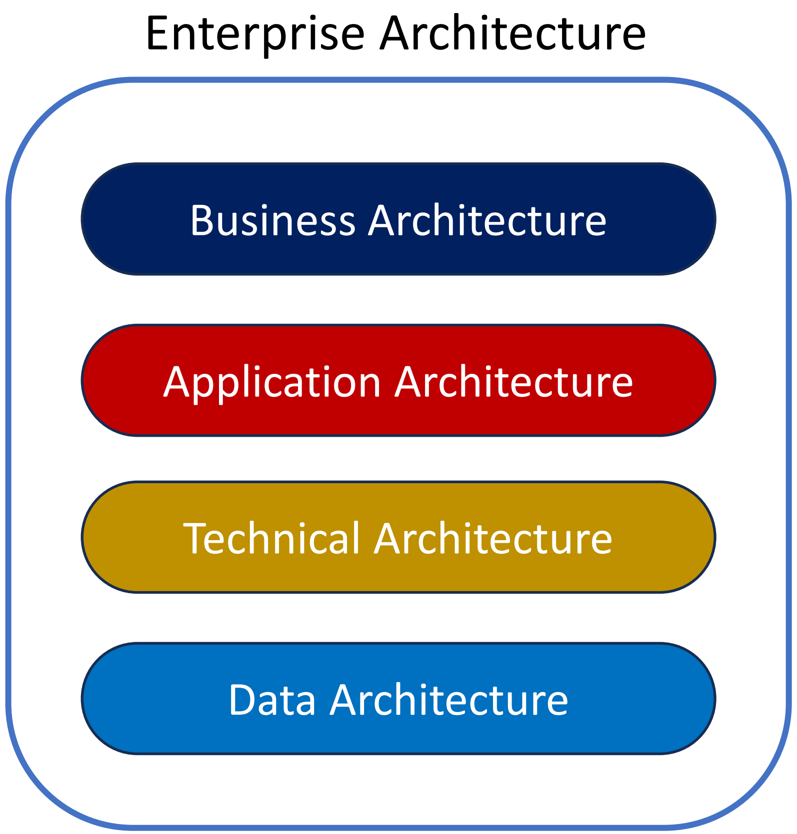
“The design of systems to **support change in an enterprise**, achieved by flexible and reversible decisions reached through careful evaluation of tradeoffs.”

“The design of systems to **support evolving data needs of an enterprise**, achieved by flexible and reversible decisions reached through careful evaluation of tradeoffs.”

Enterprise Architecture

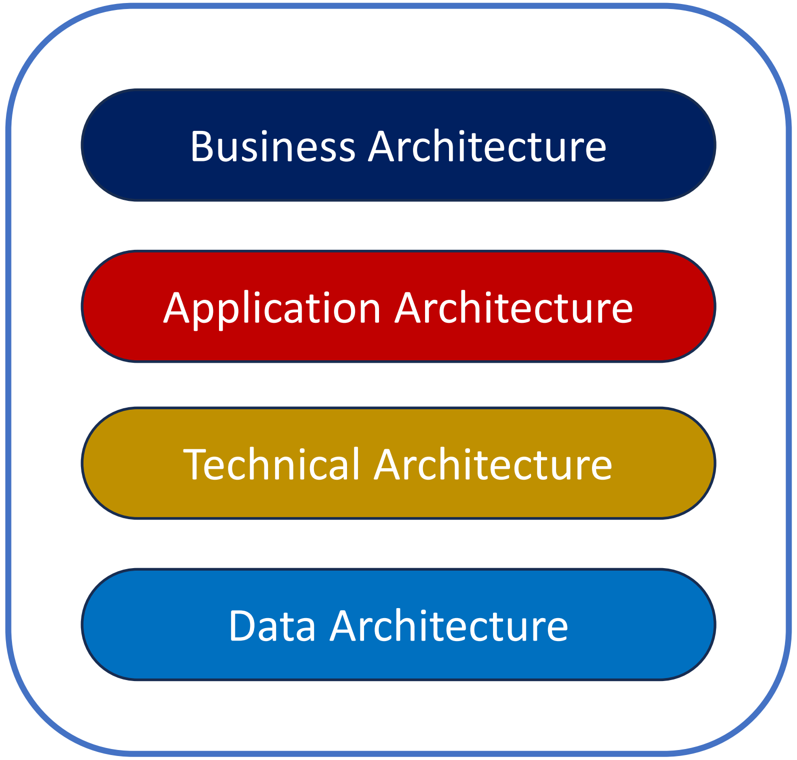


Enterprise Architecture



Enterprise Architecture

Enterprise Architecture



Product or Service strategy and business model

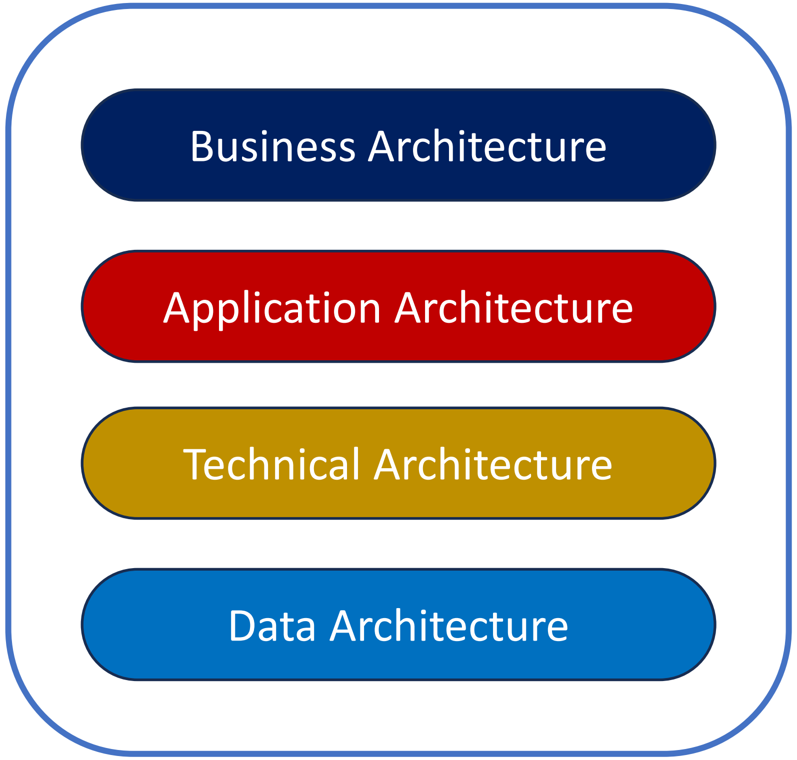
Structure and interaction of key applications

Interaction of software and hardware components

Supporting the evolving data needs

Enterprise Architecture

Enterprise Architecture



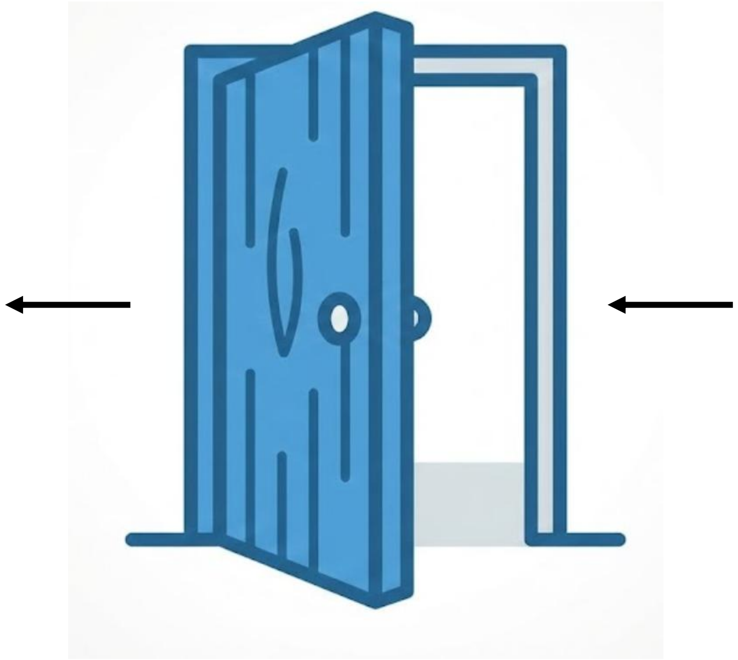
Change Management

Needs of tomorrow

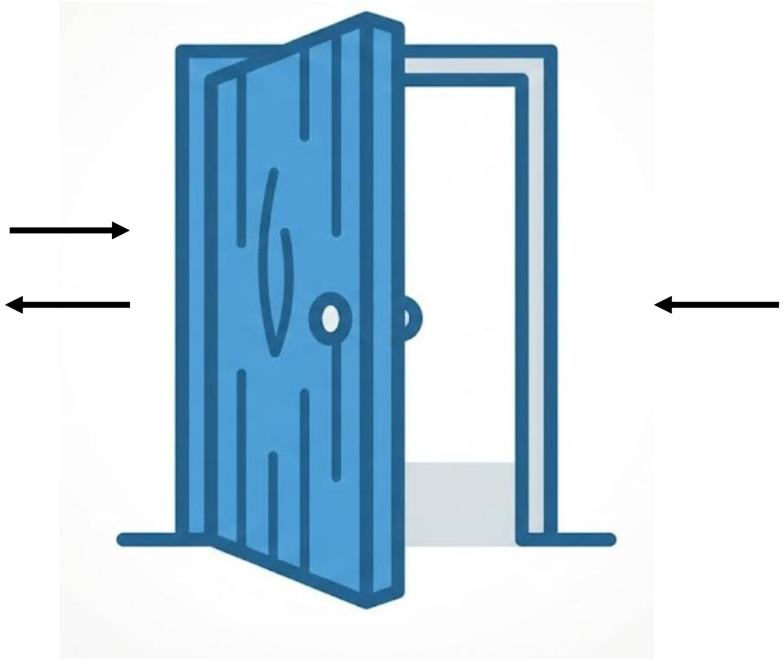
Needs of today

Adapt to organizational changes

One way and Two way door decisions

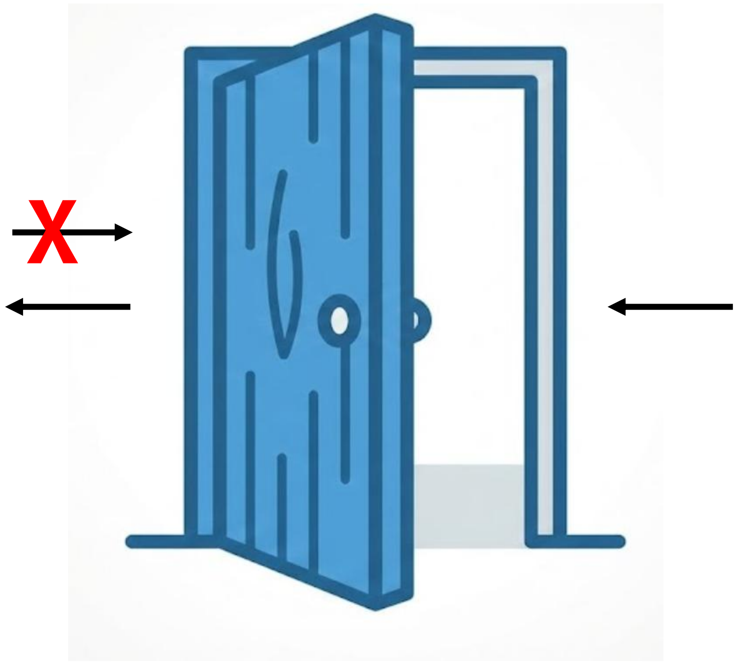


One way door decisions:
Decisions that are almost impossible to reverse



Two way door decisions:
Decisions that can be reversed

One way and Two way door decisions



One way door decisions:
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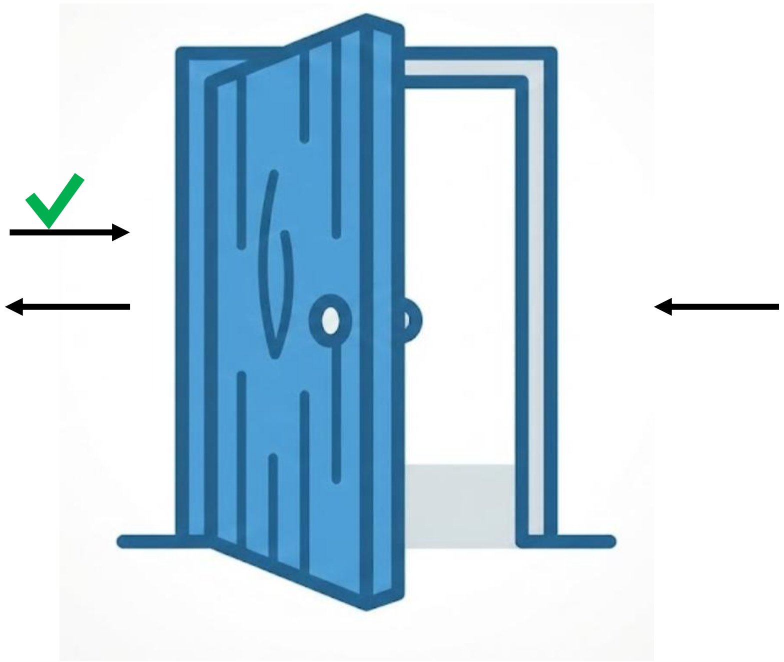


Two way door decisions:
Decisions that can be reversed

Two way door decisions

Conways Law

“Any organization that design a system will produce a design whose structure is a copy of the organization’s communication structure.”

Conways Law

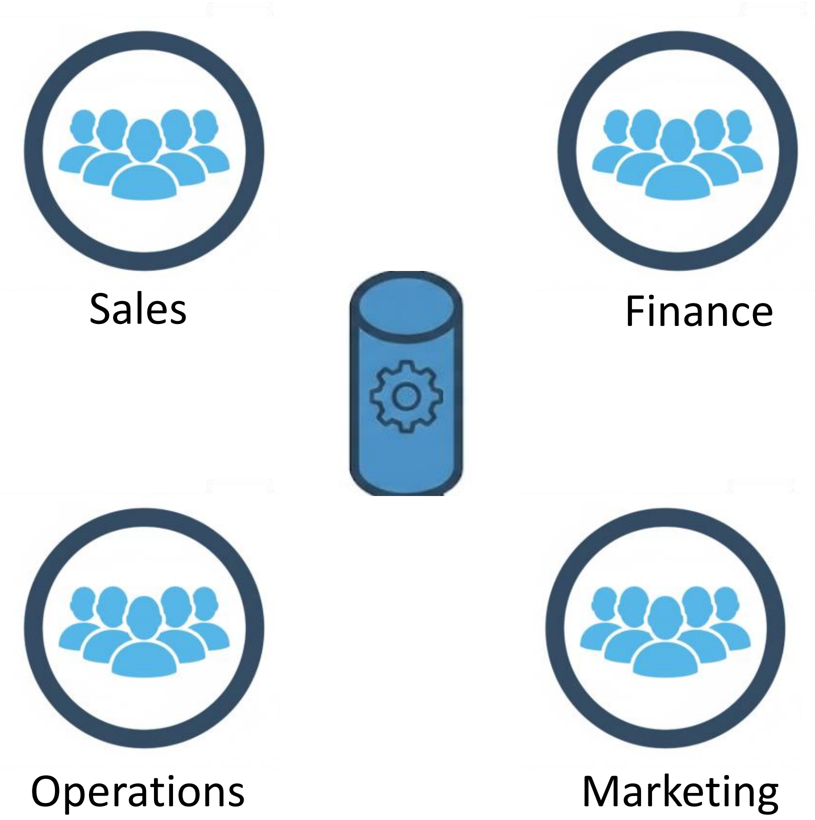
"Any organization that design a system will produce a design whose structure is a copy of the organization's communication structure."

Working in silos

Conways Law

"Any organization that design a system will produce a design whose structure is a copy of the organization's communication structure."

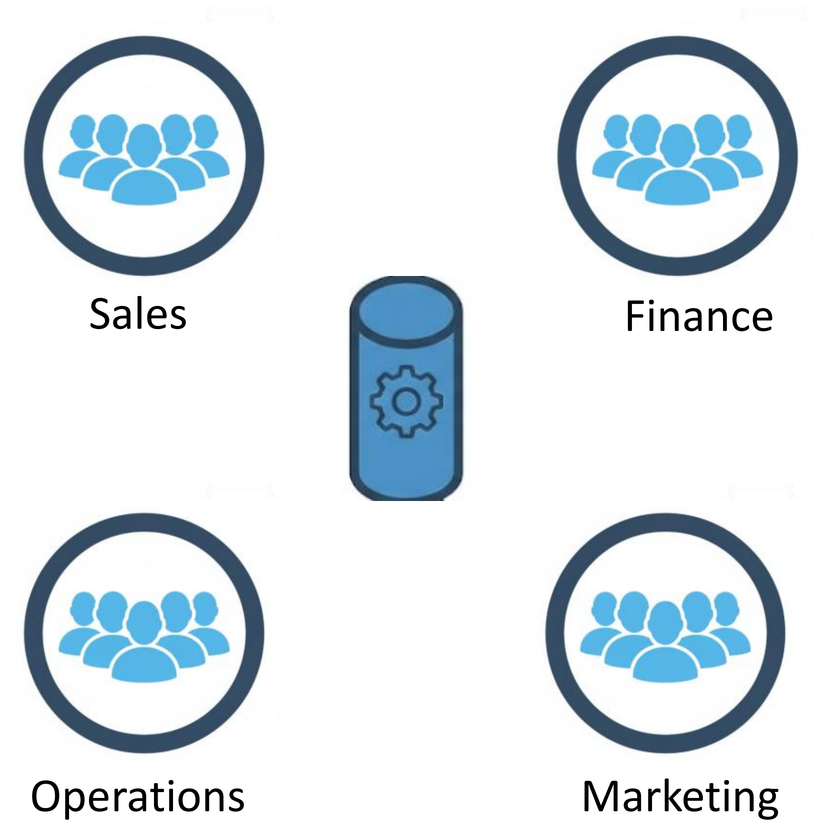
Cross functional
Communication



Conways Law

"Any organization that design a system will produce a design whose structure is a copy of the organization's communication structure."

When building data architecture we first need to pay attention to the communication structure of the organization



Principles of Good Data Architecture

- Choose common components wisely
- Architecture is leadership

- Always be architecting
- Build loosely coupled systems
- Make reversible decisions

- Plan for failure
- Architect for scalability
- Prioritize security
- Embrace FinOps

Principles of Good Data Architecture

- Choose common components wisely
- Architecture is leadership

How data architecture impacts other teams and individuals

- Always be architecting
- Build loosely coupled systems
- Make reversible decisions

Data architecture is ongoing process

- Plan for failure
- Architect for scalability
- Prioritize security
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Unspoken but understood priorities

Principles of Good Data Architecture

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Unspoken but understood priorities

Principles of Good Data Architecture

- Choose common components wisely
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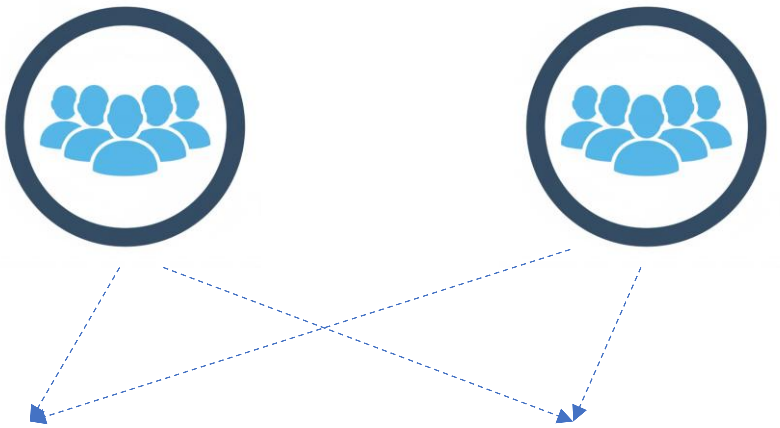
How data architecture impacts other teams and individuals

Principles of Good Data Architecture

- **Choose common components wisely**

How data architecture impacts other teams and individuals

Identifying tools and components that can benefit all teams



Common component Common component

Examples
- Object Storage
- Version control
- Processing engines